SWAPNIL DHIMAN

Senior iOS Engineer | Applied AI: LLM Agents, AI Systems and On-Device LLMs (Apple Foundation Models)

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SUMMARY

Senior iOS engineer with **4+** years at Walmart Global Tech shipping identity and commerce features in Swift to **10M+** users across the US, Mexico, Canada and Bodega. Builds AI systems end to end: LLM routing, multi-agent systems, Model Context Protocol (MCP) servers, agent skills and on-device LLMs with Apple Foundation Models. **Top 20 of 3,000+** in Walmart's Global Tech Hackathon 2026.

EXPERIENCE

Walmart Global Tech, Bengaluru **07/2022 – Present**

Senior Software Engineer, iOS (SDE III) **07/2024 – Present**

AI systems: LLM agents, model routing and on-device LLMs

- Built **WibeyConductor**, a per-turn LLM router inside Walmart's AI coding assistant: embedding-based intent classifier (e5-small-v2, **12** intents), tiered model selection and prompt-cache-aware switching in **34-61 ms** per decision. **73%** cheaper than the premium auto mode in benchmarks; part of **Token-OS** (Rust MCP server, **21** tools, **-28.5%** cost per session), a **Top 20 of 3,000+** hackathon finalist.
- Co-built **MIMO**, a macOS recorder that turns one narrated screen demo into a reusable AI agent skill; **~1K** installs on the internal registry; demoed to International Tech leadership.
- Built **Walmart Enterprise Brain** (FastAPI, Neo4j, Google Agent Development Kit), a multi-agent system that turns company documents into a causal knowledge graph: **5-stage** ingestion with embedding-based entity resolution; a founder agent and **5** lens agents answer leadership questions with cited evidence.
- Designed on-device fraud detection with an on-device LLM (**Apple Foundation Models**): tool calling over a hashed event journal, **~1.5 s** budget, **zero** inference cost; wrote the Core AI (WWDC26) proposal for in-app camera search.

iOS identity and commerce (Swift, SwiftUI, UIKit, GraphQL)

- Shipped the **Bodega identity account merge** end to end: app-launch and sign-in triggers, email step-up, merge preview and token exchange; launched in Mexico 08/2026, migrating **5.7M** overlapping identities.
- Delivered consent-based systematic merge and **Global One App account linking** (US and Mexico) with OTP and risk-based step-up, server-driven pages and analytics dashboards; both live in production.
- Implemented Native Post Purchase for Canada and Mexico, processing **1M+** transactions/month and cutting average return cycle by **25%**.
- Built Delivery Pass free trial, reactivation and membership flows for Mexico and Canada (**+25%** subscriptions in 6 months); shipped Tempo CMS plugins that let **10+** business teams update content without app releases (**-45%** release time).

Software Engineer, iOS (SDE II) **07/2022 – 07/2024**

- Built **Omni ID** for the Walmart Canada and US apps end to end: a dynamic identity QR code on the Account page that refreshes on every visit and on a GraphQL-driven expiry, with expired-state regeneration, an expanded bottom-sheet view and an onboarding journey behind a remote feature flag; raised order verification efficiency by **40%** (Bravo Award).
- Built phone-verification flows with dashboards adopted by **5+** teams; led an MVVM + Coordinator refactor (**-30%** bug rate); built web-native account deletion from scratch, with guides adopted across teams for market compliance.

Samsung R&D Institute India, Delhi | Software Development Intern **06/2021 – 07/2021**

- Built a deep neural network (T-Net and convolution layers) for LiDAR point-cloud classification (**90.88%** on ModelNet10).

PROJECTS

- **VoiceNote, Dictate into any App** (Swift, macOS): on-device speech recognition, an on-device LLM (Foundation Models) sorting into notes and todos, and OCR under the pointer, with **zero** network calls.
- **Sports AI Commentary from Two Local LLMs** (Python, IIT Kharagpur capstone): action recognition, game state, an LLM commentator and an LLM evaluator with a feedback loop, fully local.

SKILLS

iOS: Swift, SwiftUI, UIKit, MVVM + Coordinator architecture, GraphQL, REST, XCTest, Xcode, Universal Links, Passkeys, VoiceOver

AI and ML: LLM agents, multi-agent systems, Model Context Protocol (MCP), LLM routing, on-device LLMs (Apple Foundation Models), prompt caching, embeddings, Retrieval-Augmented Generation (RAG), knowledge graphs (Neo4j), Google Agent Development Kit (ADK), Core ML, TensorFlow, Hugging Face, Vertex AI

Languages and tools: Swift, Python, C++, FastAPI, Claude Agent SDK, Firebase, Git, CI/CD, Cursor

Engineering practices: feature flags, staged rollouts, A/B experiments, analytics, release management, code review, mentoring

EDUCATION

Indian Institute of Technology (BHU), Varanasi | B.Tech + M.Tech, Mathematics and Computing, CGPA **8.45/10** **2022**

M.Tech thesis: speech emotion recognition with CNN+LSTM (TensorFlow, Librosa), **87%** accuracy on RAVDESS and TESS.

IIT Kharagpur x Walmart | Applied AI training program and capstone **2026**

AWARDS

Making a Difference Award (2026) | Global Tech Hackathon Top 20 finalist (2026) | Bravo Award | Customer Champion badge